higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

AUTOMOTIVE REPAIR AND MAINTENANCE
NQF LEVEL 2

12 MARCH 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1	1.1.1	J		
	1.1.2	E		
	1.1.3	H		
	1.1.4	A		
	1.1.5	D		
	1.1.6	B		
	1.1.7	C		
	1.1.8	F		
	1.1.9	G		
	1.1.10	I		
			(10 × 1)	(10)
1.2	1.2.1	True		
	1.2.2	False		
	1.2.3	False		
	1.2.4	True		
	1.2.5	False		
			(5 × 1)	(5)
1.3	1.3.1	Safety signs		
	1.3.2	Safety shoes		
	1.3.3	Accident		
	1.3.4	Dressing		
	1.3.5	Handle		
			(5 × 1)	(5)
1.4	•Flammable liquids •Ordinary combustibles •Flammable gas •Electrical fires		(Any 2 × 1)	(2)
1.5	1.5.1	•Follow the company's rules and regulations. •Report any unsafe or unhealthy conditions. •Look after their own health and safety and that of co-workers. •Keep their workplace clean and tidy. •Always wear the required safety clothing and safety equipment.	(Any 2 × 1)	(2)
	1.5.2	Occupational health and safety		(1)
1.6	1.6.1	Ring squeezer		
	1.6.2	Outside micro-meter		
	1.6.3	Cylinder hone		
	1.6.4	Plastic gauge		
			(4 × 1)	(4)
1.7	1.7.1	1 320 millimetres		
	1.7.2	5,1 kilograms		
			(2 × 1)	(2)
				[31]

QUESTION 2

2.1
- Select the appropriate jack according to the weight of the vehicle.
- Place chocks at the rear wheels to prevent them from moving.
- Find the correct jacking point and jack up the vehicle to the required height.
- Place safety stands under the vehicle at safe points.
- Lower the vehicle slowly onto the safety stands. (Any 4 × 1) (4)

2.2 On the downward movement the piston is forced down due to the combustion pressure on the power stroke. The intake port is open and the exhaust and transfer ports are closed. While moving down, the exhaust and transfer ports are opened allowing the exhaust gasses to exit the cylinder. The downward movement also closes the inlet port. The new mixture in the crankcase is put under pressure below the piston and is forced up to the top via the transfer port. (5)

2.3 A – Rockers/rocker arm
B – Valve spring
C – Piston (3)

2.4 It keeps the flow-rate of the oil constant by preventing the oil from becoming too thin at high temperatures and too thick at low temperatures. (3)

2.5 For optimal engine function and reduced consumption, it is important for the engine to heat up as quickly as possible to operating temperature. Combustion produces heat therefore it is used to get the engine to operating temperature as quick as possible by shutting off the cooling system. (3)

2.6
- Oil pick-up strainer
- Oil filter (2)

2.7
- The mechanical fan is driven by engine power, whereas the electric fan does not put any load on the engine.
- The mechanical fan runs at engine speed and is not effective at low speeds, whereas the electric fan runs at a constant speed.
- The electric fan senses the coolant temperature directly via the thermo-fan switch which will activate the fan when cooling is needed. (Any 2 × 1) (2)

2.8 Anti-freeze has an important function. It raises the boiling point and lowers the freezing point of water. It lubricates the parts and prevents rust in the system. (2)
[24]

QUESTION 3

3.1 3.1.1
- Less impact on the suspension parts will extend their life-span
- More comfortable ride
- Better control
- More even tyre wear (Any 2 × 1) (2)

3.1.2
- Fit lighter wheel rims
- Fit lighter control arms
- Fit lighter brake discs
- Reduce the stub-axle weight (Any 2 × 1) (2)

3.2 3.2.1
A – Tapered roller bearing
B – Needle roller bearing (2)

3.2.2
- Needle roller bearings can be fitted in narrow spaces.
- Needle roller bearings are made in a bigger variety of shapes and sizes. (Any ONE) (1)

3.3 They are torqued according to specification. (1)

3.4
A – Over-inflated wheels
B – Under-inflated wheel
C – Incorrect wheel alignment (3)

3.5
- The physical size of the battery. It must fit in the battery tray.
- The battery should not be too high that the bonnet will touch the posts.
- The amp/hour rating of the battery is correct.
- The posts should be the same as the old battery otherwise the cables might not reach. (Any 2 × 1) (2)

3.6
- The battery posts
- The casing
- The vent plugs
- The filler holes (Any 3 × 1) (3)

3.7 2,1 Volts or 2 Volts (1)

3.8
3.8.1 1230–1250
3.8.2 1120–1150
3.8.3 1280–1350
(3 × 1) (3)

3.9
- Lens
- Reflector
- Adjusters
- Housing
- Bulb (4)

3.10 Light emitting diode (1)
[25]

QUESTION 4

4.1
- It shows the vehicle's detail to confirm you are working on the correct vehicle
- It lists the customer's request for the work they want done.
- It shows the list of parts the mechanic fitted.
- It shows the work done by the mechanic for record keeping.
- It is used to determine the amount the customer must pay. (Any 2 × 1) (2)

4.2
- Engine oil leaks
- Coolant leaks
- Brake fluid leaks
- Transmission fluid leaks
- Steering fluid leaks (Any 3 × 1) (3)

4.3
- A hoist
- Compressor
- Tyre pressure gauge
- Timing light
- Oil drain pan (Any 2 × 1) (2)

4.4
- Engine oil
- Gearbox oil
- Differential oil
- Power steering oil (Any 3 × 1) (3)

4.5
- Park lights
- Headlights on dipped beam
- Headlight on bright
- Hazards
- Indicators
- Fog lights
- Spotlights (Any 4 × 1) (4)

4.6
- Dirty rags/cloth
- Old metal parts
- Parts packaging (cardboard boxes, plastic bags)
- Old oil (Any 3 × 1) (3)

4.7
- To check whether the job was done correctly.
- To check that no faults were created during the work done.
- To check whether the vehicle is safe for the customer to use.
- To check the vehicle is performing like it should. (Any 3 × 1) (3)

[20]

TOTAL: 100

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